



LG G2

LG set to reveal its new flagship, the G2, on Aug 7 in New York; drops the Optimus tag. <http://dgit.in/12eiz7X>



Google's gaming console?

Google might be jumping the bandwagon and is reportedly designing an Android based gaming console. <http://dgit.in/1aTKxXI>

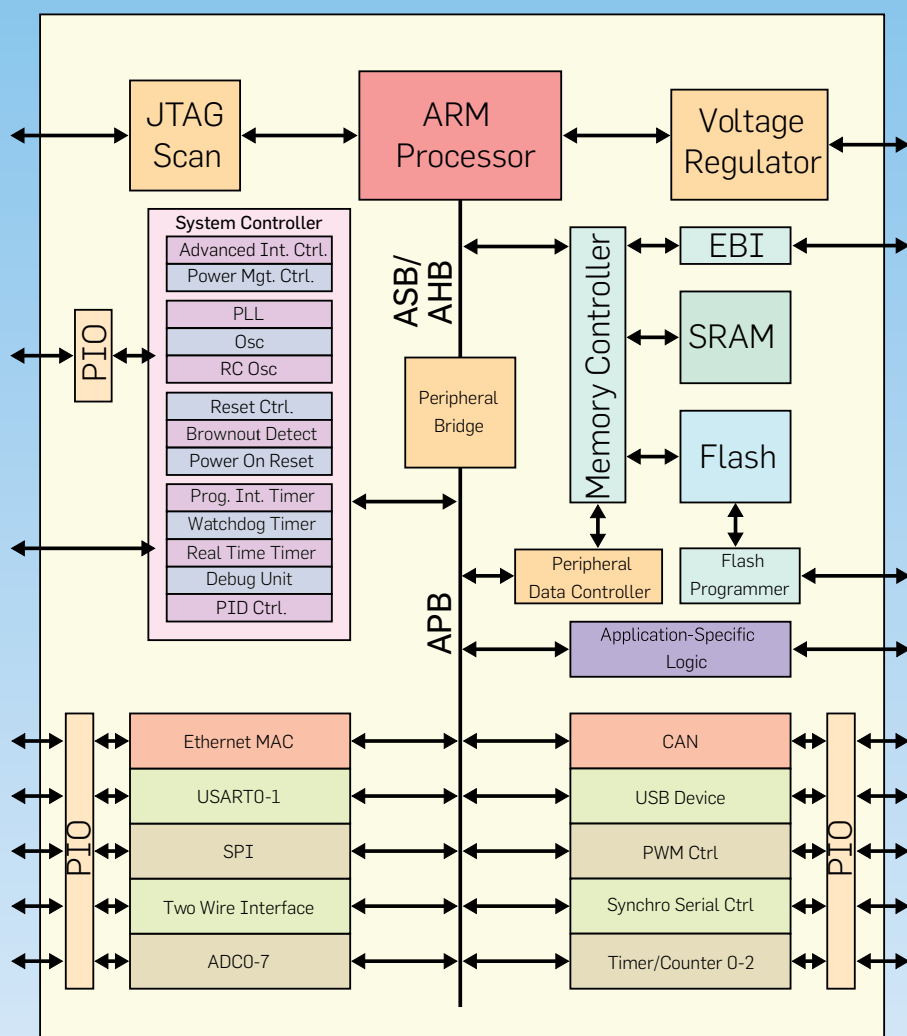
SoC 101

It's all about taking the essence of the biggest, baddest computer ever built complete with all its myriad components and shrinking it down to one piece of silicon die. In a nutshell, that's what an SoC is. Honey, they shrunk the chip!

After making that spectacular announcement, let's take a closer look at what an SoC or System-on-Chip is. The SoC is basically an integrated circuit, with all the parts of a computer on one single chip. It contains within it the processor cores, the graphics chips, the cellular radios, mobile RAM, image processors, audio/video decoders and so on. On a typical PC motherboard, all the different components are housed in separate chips – digital, analog, mixed-signal, etc. – including the main CPU. However, on an SoC all of these separate components are fused together on a single chip, and the resulting SoC is hardly bigger than a traditional PC processor.

All separate components in their own right, but shrunk down and consolidated onto a single silicon die for specialist devices. What kind of devices, we hear you ask? Anything that sports an embedded system, but more popularly on smartphones and tablets. The consolidation of all these seemingly separate functional elements on a single chip also helps in keeping the power requirements of an SoC substantially lower than all of these individual components when connected in a complex system – say, a desktop computer.

Another thing to remember: while a desktop is large, an SoC-based smartphone is tiny and with plenty of room for batteries. And if you think an SoC is just a truncated version of a larger, more complex computer system, and equally slower in performance, you couldn't be more wrong. NVIDIA Tegra and Qualcomm's Snapdragon-based SoCs are nippy and give budget laptops a good run for their money in terms of hardware capabilities.



This block diagram of an ARM SoC illustrates how everything sits on a single silicon die

Just like Intel rules the roost in the PC chip market, ARM (A British company) is the undisputed king among SoC makers. It makes the reference architecture design based on RISC (reduced instruction set computing) and licenses it out to hardware manufacturers like Qualcomm, Samsung, NVIDIA, and so on, who in turn add on to ARM's good work by tweaking, perfecting these designs for their needs. Think of it as Google designing Android and sharing it with HTC and other phone manufacturers, wherein these phone manufacturers tweak Android as

per their needs (HTC Sense, etc.) and deploy it on their devices.

Each and every SoC manufacturer has its own unique nomenclature for their tweaked SoCs. NVIDIA calls theirs Tegra, whereas Qualcomm calls theirs Snapdragon and Krait. Texas Instruments makes OMAP, while Broadcom keeps launching a new one under their BCMxxxx moniker. Unlike Intel which makes PC chips, there are many SoC manufacturers, and all their SoCs have something different than the other – they have to be, otherwise who would buy them?